

Assignment – 17:

Problem Statement:

To accept the number of terms and find the sum of the sine series.

Code:

```
#include <stdio.h>

int main()
{
    int n, i, j;
    float x, term, sum = 0;
    printf("Enter value of x (in radians): ");
    scanf("%f", &x);
    printf("Enter number of terms: ");
    scanf("%d", &n);
    for(i = 1; i <= n; i++)
    {
        term = 1;
        for(j = 1; j <= (2*i - 1); j++)
        {
            term = term * x;
        }
        for(j = 1; j <= (2*i - 1); j++)
        {
            term = term / j;
        }
        if(i % 2 == 0)
        {
            term = -term;
        }
        sum = sum + term;
    }
}
```

```
}  
printf("Sum of sine series = %f\n", sum);  
}
```

Input & Output:

```
Enter value of x (in radians): 10  
Enter number of terms: 5  
Sum of sine series = 1448.271729
```

```
Enter value of x (in radians): 1  
Enter number of terms: 5  
Sum of sine series = 0.841471
```
